21CSE355T / Data Mining and Analytics

Question Bank

Unit-1

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1. Explain different kinds of data used in data mining with relevant examples.

2. Identify and describe different kinds of patterns that can be discovered in data mining. Provide examples for each.

3. Analyze the different types of data objects and attribute types used in data mining. How do they influence the choice of data mining techniques?

4. Design a data visualization plan to analyze a dataset containing sales data over time. Discuss the techniques you would use and their benefits.

5. Evaluate the importance of data cleaning in data preprocessing. What are the common techniques used for data cleaning?

6. How is data integration performed in data preprocessing? Discuss its significance and methods.

7. Analyze the process of attribute subset selection. Why is it important in data preprocessing, and what techniques are used?

8. Explain the concepts of clustering and sampling in data preprocessing. Provide examples of their applications.

9. Identify and describe different kinds of patterns that can be discovered in data mining. Provide examples for each.

10. Explain the concepts of clustering and sampling in data preprocessing. Provide examples of their applications.

11. Given a dataset from a smart home IoT system containing records such as device ID, device type, usage frequency, energy consumption, and environmental variables (e.g., temperature, humidity), apply the data pre-processing steps you would take to prepare the data for a mining process and explain each step with practical examples from the dataset.

12.Given a dataset from a healthcare system that includes patient records with fields such as patient ID, age, gender, diagnosis, treatment, and medical history, apply the data pre-processing steps you would take to prepare the data for a mining process and Explain each step with practical examples from the dataset.

Unit-2

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1. Explain the concept of Market Basket Analysis and its applications in data mining. Provide examples.

2. Describe the Apriori algorithm and its process for mining frequent itemsets. Provide an example to illustrate the steps.

3. Analyze the differences between mining frequent itemsets using traditional methods (like Apriori) and methods that do not generate candidates (like FP-Growth).

4. Design a strategy for mining frequent itemsets using the vertical data format. Discuss its advantages over the horizontal data format.

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5. Evaluate the importance of mining closed frequent itemsets. How do they differ from frequent itemsets?

6. Describe the concept of mining multilevel association rules. How does it differ from mining single-level association rules?

7. Explain the role of correlation analysis in data mining. How is it used to enhance association rule mining?

8. Discuss the concept of constraint-based association mining. Design a scenario where applying constraints can improve the efficiency and relevance of mining results.

9. Analyze the differences between mining frequent itemsets using traditional methods and methods that do not generate candidates.

10. Explain the role of correlation analysis in data mining. How is it used to enhance association rule mining?

11. Generate Frequent patterns for the following transactions using Apriori algorithm for the given scenario:
A supermarket wants to analyze customer purchasing patterns to optimize product placements and cross-promotions. Given Minimum Support=40% and Confidence=70% Generate Association Rules: From the frequent itemsets, derive rules that meet the minimum confidence level. Given Transactions are

Transaction ID	Itemsets
1	{ Bread, Butter, Milk }
2	{ Bread, Butter }
3	{ Beer, Cookies, Diapers }
4	{ Milk, Diapers, Bread, Butter }
5	{ Beer, Diapers }

12. Generate Frequent patterns for the following transactions using Apriori algorithm for the given scenario: A supermarket wants to analyze customer purchasing patterns to optimize product placements and cross-promotions.

a. Given Minimum Support>30% and Confidence=60%
b. Generate Association Rules: From the frequent itemsets, derive rules that meet the minimum confidence level.
c. Given Transactions

Transaction ID	Itemsets
1	{ Bread, Milk }
2	{ Bread, Butter, Juice, Eggs }
3	{ Milk, Butter, Juice, Cola }
4	{ Bread, Milk, Butter, Juice }
5	{ Bread, Milk, Cola }
6	{ Milk, Butter, Cola }
7	{ Bread, Butter }
8	{ Juice, Butter }
9	{ Bread, Milk, Juice }
10	{ Milk, Eggs }

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